Type MS · State what antibiotics treat and explain how resistance develops

Answer 1 of 9 · Question 1 part b · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

<table>
<tr>
<td>1 (a)</td>
<td>
<table>
<tr><th>Name of disease</th><th>Type of organism</th><th>Method of transmission</th></tr>
<tr><td>malaria</td><td>protozoan;</td><td>mosquito;</td></tr>
<tr><td>poliomyelitis</td><td>virus;</td><td>air/water;</td></tr>
<tr><td>typhoid</td><td>bacterium;</td><td>food/water/housefly;</td></tr>
</table>
</td>
<td>For methods of transmission:

Allow anopheles/vector

Allow droplets for water

Allow vector for housefly</td>
<td>6</td>
</tr>
<tr>
<td>(b)</td>
<td>• caused by a fungus (1)
• (antibiotics) only effective against bacteria/ not effective against fungi (1)</td>
<td>Allow (antibiotics) do not kill fungi</td>
<td>2</td>
</tr>
</table>

Total 8 marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 1 of 9 · Question 1 part b · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 2 of 9 · Question 6 part b.i · 1 mark

2020 JAN · Paper 02 · find part b.i in this mark scheme extract

Question number					
6	(a)	(i)	• more males than females die (from *S.aureus* and MRSA)(1) • number of deaths from both *S.aureus* and MRSA increase with age(1)	Allow other valid conclusions e.g more deaths from both microorganisms in males/females aged 85 and over	2
		(ii)	416.6 ÷ 1.8(1) 231.4:1 (1)	Allow 231:1 Allow full marks for correct final answer	2
		(iii)	41.2 + 21.9 = 63.1(1) 53 × 63.1 = 3344(1)	Allow 3344.3 Allow full marks for correct final answer	2
		(iv)	Any two from: • MRSA is resistant (1) • to (many) antibiotics(1) • (MRSA infection) difficult to treat(1) • reference to MRSA infection being (highly) contagious/out of control/easily transferred(1)		2
	(b)	(i)	mircroorganisms/bacteria/fungi	Ignore viruses Ignore named antibiotics	1
		(ii)	Any six from: • sterilise Petri dishes/agar/nutrient broth(1) • obtain sample(1) • sterile inoculating loop/sterile swab(1) • pass neck of flask containing culture through a flame(1) • smear culture/sample/swab onto agar(1) • reference to adding antibiotics (to agar)(1) • seal Petri dishes/cover with lid(1) • incubate(1) • measure area of clear zone around antibiotic discs(1)	Allow description of how loop/swab is sterilised	6

Total for Question 6 = 15 marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 3 of 9 · Question 7 part a.ii · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
7				
(a)	(i)	C; (pathogen) A not all bacteria cause disease B not all fungi cause disease D not all protozoa cause disease		1
	(ii)	• Ebola caused by a virus; • antibiotics only affect bacteria/not active against viruses;		2
(b)	(i)	• 700 - 180 = 520; • 520 ÷ 5 (= 104); • 100 (AU per hour);	Ecf max 2 marks if first mp incorrect but value divided by 5 to arrive at correct answer Max 1 if first answer correct but division is incorrect	3
	(ii)	• B is bactericidal; • because population decreases/kills bacteria;		2
	(iii)	• antibiotic B; • because it kills bacteria/causes population to decrease; • but symptoms may be less severe/mild/reduced; • antibiotic A leaves large numbers alive/bacterial population remains constant/only stops/reduces population growth; • these can still cause disease;		3 Total 11

Type MS · State what antibiotics treat and explain how resistance develops

Answer 3 of 9 · Question 7 part a.ii · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 4 of 9 · Question 8 part b.i · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

8	(a)	(i)	A = flagellum; B = cell wall/capsule; C = nucleic acid/genetic material/bacterial chromosome/(loop of) DNA;		3 (grad)
		(ii)	pathogen;		1 (cler)
	(b)	(i)	• produced by microorganism/bacteria/fungi; • kills bacteria;		2 (grad)
		(ii)	A *no mark for giving X* • because larger clear zone; • antibiotic diffused from filter paper; • bacteria have been killed;		3 (exp)
		(iii)	• filter paper without antibiotic; • bacterial growth unaffected/no clear zones; • if antibiotic is causing effect;		3 (exp)

Total 12 marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 4 of 9 · Question 8 part b.i · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

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Type MS · State what antibiotics treat and explain how resistance develops

Answer 5 of 9 · Question 4 part b.i · 1 mark

2024 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

4 (a)	• diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly; • malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ;	Ignore ORT	2
(b) (i)	antibiotics;		1
(ii)	• elderly have weakened/lowered immune system/immunocompromised; • indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together;		2
(c) (i)	• diet; • exercise; • smoking: • drinking alcohol: • stress	Accept drugs	2
(ii)	• 5 000 000 / 100 000 = 50; • 50 × 445 = 22 250; • 22 300; ***22,250 scores 2 marks*** ***22,300 scores 3 marks***	***1 mark for ×445 if there are no other marks***	3
(iii)	• use of 445 and 185; • 445 / 185 ; • 2.41:1/2.405:!/2.4:1= 3 marks;	ecf from mp1 **if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks	3
(iv)	• people live longer/elderly population; • these diseases are more common in older people; • less likely to die younger from infectious diseases/better medical facilities/treatment; • different diet/ more meat/saturated fats/processed food; • drinking more alcohol;	People live longer so there is more cancer and dementia/these diseases= 2 marks	2

Total 15 marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 5 of 9 · Question 4 part b.i · 1 mark

2024 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 6 of 9 · Question 6 part a.iv · 2 marks

2020 JAN · Paper 02 · find part a.iv in this mark scheme extract

Question number				
6	(a) (i)	• more males than females die (from *S.aureus* and MRSA)(1) • number of deaths from both *S.aureus* and MRSA increase with age(1)	Allow other valid conclusions e.g more deaths from both microorganisms in males/females aged 85 and over	2
	(ii)	416.6 ÷ 1.8(1) 231.4:1 (1)	Allow 231:1 Allow full marks for correct final answer	2
	(iii)	41.2 + 21.9 = 63.1(1) 53 × 63.1 = 3344(1)	Allow 3344.3 Allow full marks for correct final answer	2
	(iv)	Any two from: • MRSA is resistant (1) • to (many) antibiotics(1) • (MRSA infection) difficult to treat(1) • reference to MRSA infection being (highly) contagious/out of control/easily transferred(1)		2
	(b) (i)	mircroorganisms/bacteria/fungi	Ignore viruses Ignore named antibiotics	1
	(ii)	Any six from: • sterilise Petri dishes/agar/nutrient broth(1) • obtain sample(1) • sterile inoculating loop/sterile swab(1) • pass neck of flask containing culture through a flame(1) • smear culture/sample/swab onto agar(1) • reference to adding antibiotics (to agar)(1) • seal Petri dishes/cover with lid(1) • incubate(1) • measure area of clear zone around antibiotic discs(1)	Allow description of how loop/swab is sterilised	6

Total for Question 6 = 15 marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 7 of 9 · Question 4 part c.ii · 5 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

Number					
4	(a)	(i)	cell wall/capsule/slime coat;		1
		(ii)	55 ÷ 30 000 = 0.0018: 1.8/1.83/1.833;		2
	(b)	(i)	more densely/higher populated/more people/overcrowded/more waste/rubbish/sewage;		1
		(ii)	• fly lands on faeces (of person with a stomach ulcer); • transfers the bacteria to food; • person ingests the contaminated food;		3
	(c)	(i)	Two from: • high/medium levels in less well developed/poorer countries; • where sewage treatment is less effective/water not clean/poor sanitation; • more densely/highly populated; • healthcare/vaccines to treat infected people is less effective/available;	accept reverse arguments	2
		(ii)	Five from: • bacteria mutate/acquire resistant plasmids; • people do not finish the course/over-use of antibiotics; • resistant bacteria survive (and non-resistant bacteria die); • (resistant bacteria) reproduce/multiply; • offspring are antibiotic resistant/not affected by antibiotics; • antibiotic that will be effective is needed/used; • more serious infection occurs;		5

Total 14 marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 7 of 9 · Question 4 part c.ii · 5 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 8 of 9 · Question 5 part a.iv · 4 marks

2024 JUN · Paper 02 · find part a.iv in this mark scheme extract

Number				
5	(a) (i)	**C** the type of antibiotic added The following are incorrect because: A the mass of bacteria added is irrelevant B the size of the inhibition zone is the dependent variable D the mass of agar in the Petri dish is irrelevant		(1)
	(ii)	Any two from: • sterilise equipment/named equipment/disinfect/swab desk; • use of sterile agar; • reference to use of lid on Petri dish/ minimising the time the lid is removed; • Flame from Bunsen burner:		(2)
	(iii)	(antibiotic) Q; greater inhibition zone /greater clear zone/ more bacteria killed;		(2)
	(iv)	Any four from: • mutation; • (resistant) bacteria survive; • reproduce; • pass on resistant gene/allele; • Increase of resistant gene/allele in the population/ increase in allele frequency (over many generations)		(4)
	(b)	• (bacteria/fungi/decomposers) release enzymes/named enzyme • breakdown dead/organic matter/ named organic matter; • release nutrients/named nutrients into the soil/environment;	Ignore carbon dioxide Ignore fertiliser	(2)

Total for Question 5 = 11 marks

Type MS · State what antibiotics treat and explain how resistance develops

Answer 9 of 9 · Question 8 part b.iii · 2 marks

2025 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

8 (a)	(i)	cilia/skin/mucus	allow other suitable answers	1
	(ii)	kills/inhibits growth of pathogens/damage cell membranes;		1
	(iii)	• (phagocytes) engulf pathogens/microorganisms / bacteria; • digest/breakdown pathogens/microorganisms / bacteria;		2
	(iv)	antigen/immunogen;		1
	(v)	• antibodies have a particular shape /are complementary (to antigen); • only recognise/attach/bind to, one particular/specific pathogen / antigen/receptor;		2
	(vi)	• natural passive immunity the body receives antibodies naturally;	accept through breast milk/placenta	
		• artificial passive immunity are when antibodies are injected into the body;	accept e.g. to give immediate protection against a disease;	2
(b)	(i)	360 ÷ 30 = 12; 108 ÷ 12; 9;		3
	(ii)	reduce the risk of disease transmission/infection;		1
	(iii)	any two from • improper/overuse of antibiotics/over prescribed; • bacterial DNA mutates/bacteria adapt; • some survive and reproduce		2
			Total for Question 8 = 15 marks	